

Technical Assessment for Data Scientist Position

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Armani Middle East Trading | Data engineering · Forecasting · Dashboard · Dynamic pricing

Four parts, one chain: each answer feeds the next

Part	Business problem	The question I asked first	What I built
1 Data and ETL	Raw weekly file of 30 products	Can I trust every number, and what does each column really measure?	Validated pipeline, star schema, audit trail
2 Forecast	Demand volatility	What exactly should I forecast, and how do I prove a model works?	Demand rebuild, 13-model test, 26-week forecast
3 Dashboard	Managers need KPIs and alerts	What would a manager decide on Monday morning?	Plain-language dashboard with alerts
4 Pricing and risk	Static prices, credit and liquidity	Where does price really earn money, and what could go wrong?	Rule engine, safety limits, pilot plan

Diagnose before modelling

What the data measures is understood before a method is chosen.

Test on unseen data

Every choice is judged on weeks the model has not seen.

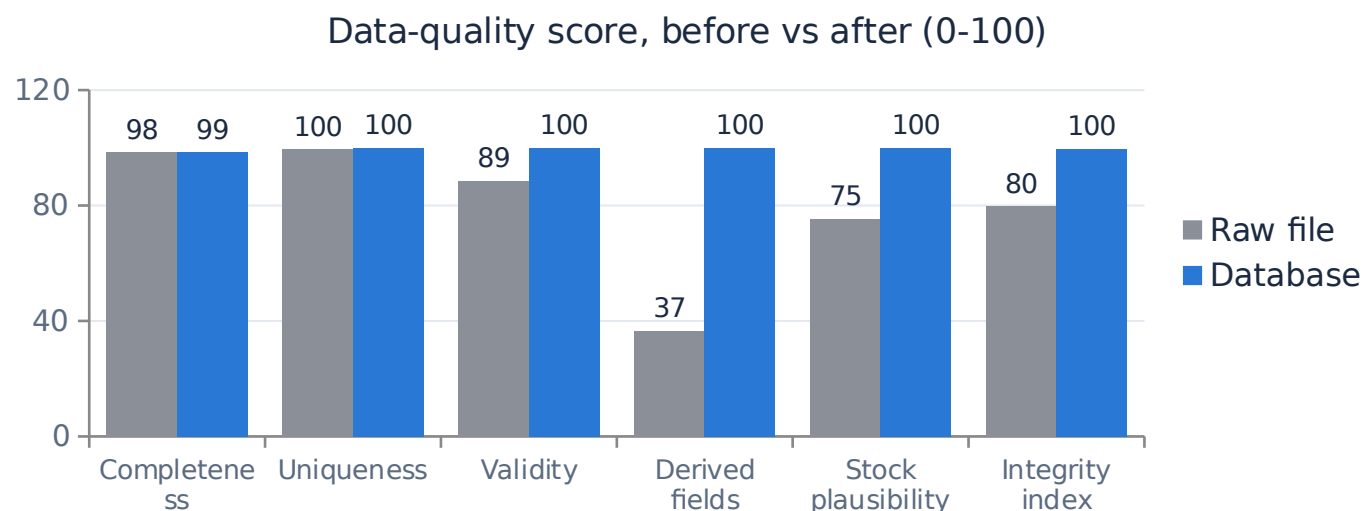
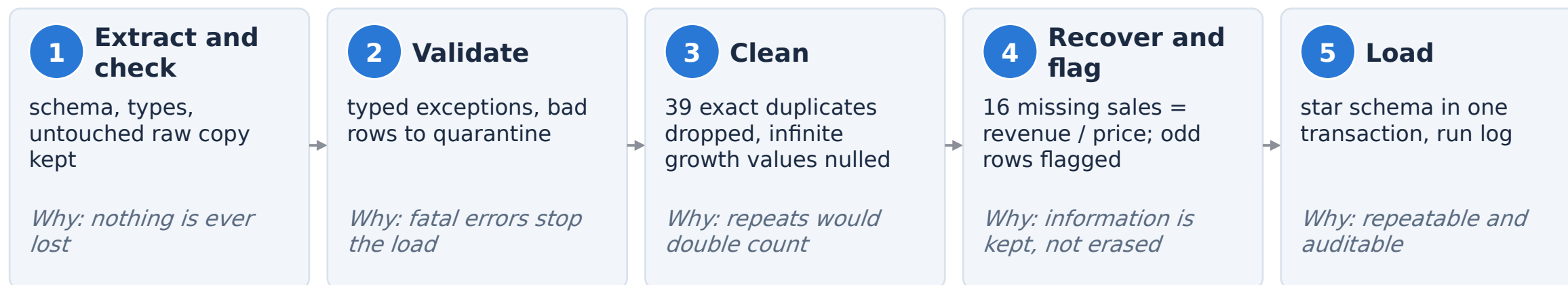
Simple unless proven otherwise

A complex model must beat a plain rule to be used.

State and stress assumptions

Where data is missing, assume openly and show the range.

A pipeline that never loses or hides information



7,800

unique product-weeks from 7,839 raw rows

15 / 15

exception tests passed on a corrupted copy

Star schema: product, date and weekly-sales tables plus audit tables, so every later question is a simple query.

Nine data problems, each treated with a reason

39

duplicate rows (Week, Product)

DROPPED

a repeat double counts sales

16

missing Sales_Units

RECOVERED

Revenue / Price_Per_Unit

606

missing Demand_Forecast

KEPT EMPTY

imputing would invent a baseline

897

infinite Revenue_Growth

RECOMPUTED

growth from zero is undefined

37%

of derived columns match (Revenue, Gross_Profit)

RECOMPUTED

KPIs must reproduce 100%

907

rows: Inventory_Level = 0 but Sales_Units > 0

REALIGNED

stock = previous row's Inventory_Level

1,329

Sales_Units = 0 weeks; pile-ups at 1,500 / 2,500

FLAGGED

stock caps, not errors

23

Sales_Units outliers

FLAGGED

winsorised copy, original kept

3

missing sources: dates, customers, competitors

ASSUMED

synthetic calendar, labelled scenarios

Data finding and the demand_est computation

0.85

correlation of sales with the PREVIOUS row's stock

1,329

stock-out weeks, every one with zero sales

2,813

weeks where sales equal stock (pile-ups at 1,500 / 2,500)

sales = min(demand, stock) → 53% of weeks understate demand → demand_est is computed per week type

Unconstrained · 46.5%
demand_est = sales

stock was enough, so sales are the demand

Barley, week 28

Sales	2,201
Stock available	2,500
Legacy forecast	2,211
Seasonal model	2,343

demand_est = 2,201observed

Stock-limited · 36.1%
demand_est = max(sales, 0.40 × legacy + 0.60 × model)

sales hit the stock level, so demand is at least that

Barley, week 2

Sales	2,500 (= stock)
Legacy forecast	2,733
Seasonal model	2,611
Blend 0.40 / 0.60	2,659

demand_est = 2,659+6.4% above sales

Stock-out · 17.0%
demand_est = max(sales, 0.40 × legacy + 0.60 × model)

no stock, so sales are 0 and demand is hidden

Barley, week 4

Sales	0 (stock = 0)
Legacy forecast	3,195
Seasonal model	2,758
Blend 0.40 / 0.60	2,925

demand_est = 2,925sales were 0

Why 0.40 / 0.60: on weeks with known demand the blend errs by 5.0%, against 7.9% for the legacy forecast and 6.6% for the seasonal model alone; weights fitted out-of-fold. 606 missing legacy values use the model alone.

Feature engineering in three layers

A Data and stock features *built in ETL*

- stock available = previous row's inventory (lag 1)
- censor status and flags: stock-out, capped, pile-up, outlier
- recomputed margin, growth, turnover

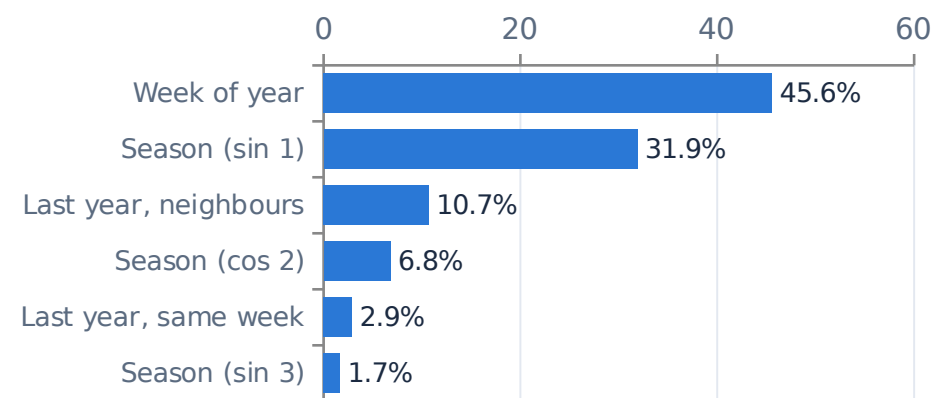
B Target and structure *built in pre-processing*

- demand_est: the reconstructed target
- calendar: week of year, quarter, season, year index
- stock regime (k-means) and high / normal stock profile

C Model inputs *built in forecasting*

- Structural: product level, shared trend, sin / cos of week
- LightGBM: horizon, same week last year (+/- 1), 13 / 26 / 52-week levels
- target relative to the 52-week level

What the learner relies on (share of importance)



78-86%

variance by
week of year

0.97

product
profile
correlation

-0.017

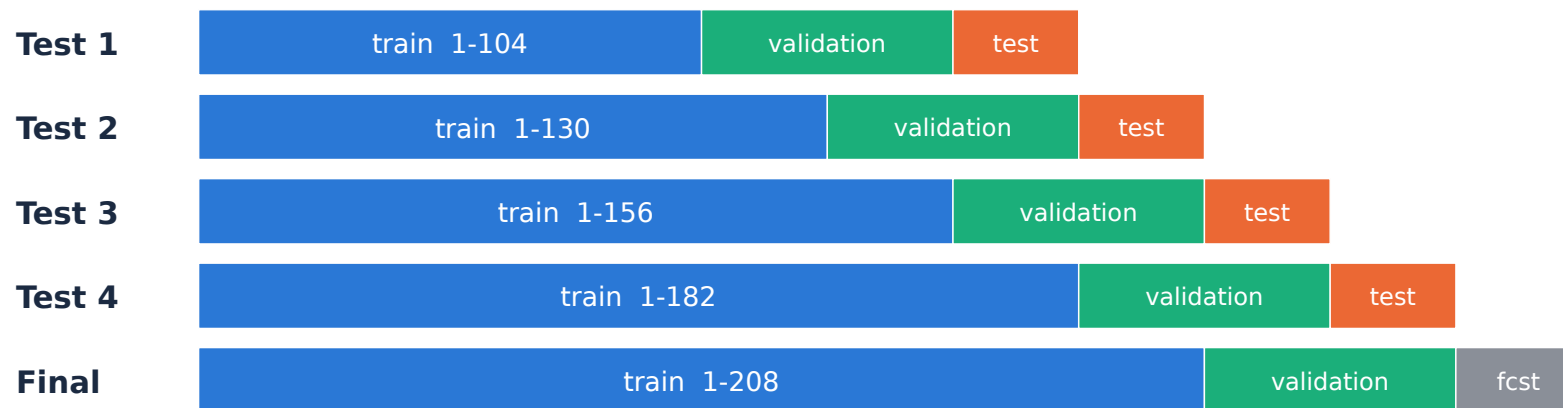
price effect
(n.s.)

Rules and decisions

- Only information known at the forecast origin is used; the legacy forecast builds the target but is never an input. One shared seasonal curve; price, cost and stock are not features.

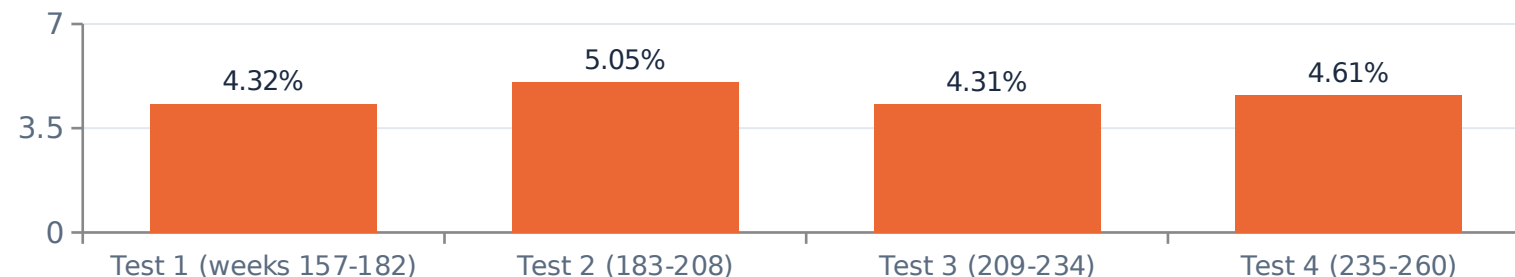
Why four rolling tests instead of one train/test split

Predicts **demand_est** (units customers wanted) · 26 weeks × 30 products = 780 forecasts · inputs: past demand, week of year, product



train = learns · validation = chooses settings · test = scores (weeks 157-182, 183-208, 209-234, 235-260);
Final: weeks 261-286

Winner's error in each test: stable between 4.3% and 5.1%



One split = one lucky number

A single test window can fall in an easy or a hard season.

Seasons alternate

Tests cover weeks 1-26 and 27-52 of the year, twice each: peak and lull both scored.

No random split

Shuffled weeks leak neighbouring weeks and the seasonal pattern into training.

Mimics real use

Always: learn from the past, forecast 26 weeks. The final forecast works the same way.

The winner: level × growth × season

$$\text{demand}(\text{product, week}) = \text{product level} \times \text{growth trend} \times \text{seasonal factor}$$

Product level *how big the product normally is*

One number per product: its average weekly demand (30 numbers).

Found because products differ 5.5x in size: 555 units a week (Wheat) to 3,054 (Tea).

Growth trend *is the business slowly growing?*

One shared yearly slope for all products: about +2.8% a year.

Found because demand grew 2.8% (last 52 vs previous 52 weeks); trend strength 0.92 in an STL decomposition.

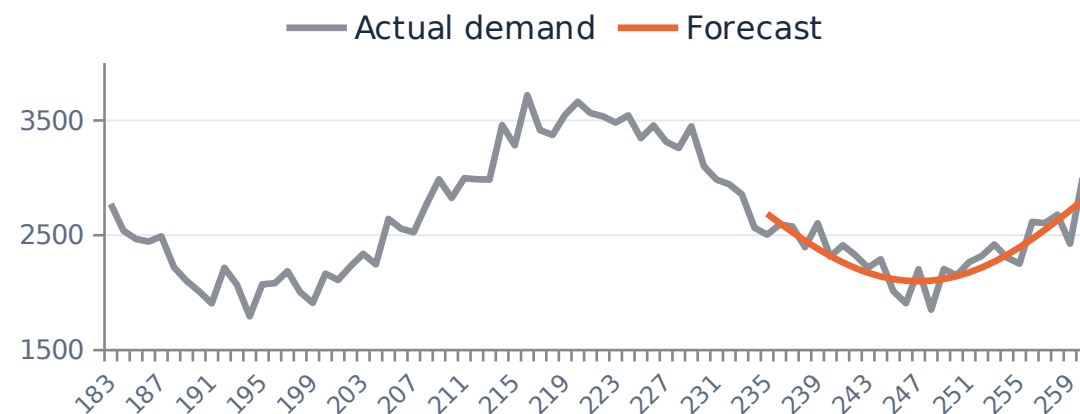
Seasonal factor *is it a busy or quiet time of year?*

A multiplier per week of the year: 1.00 normal, 1.25 = 25% busier. Same curve for all products.

Found because week of year explains 78-86% of demand variance and 30 products correlate 0.97: peak week 13 (+25%), quiet week 38 (-25%).

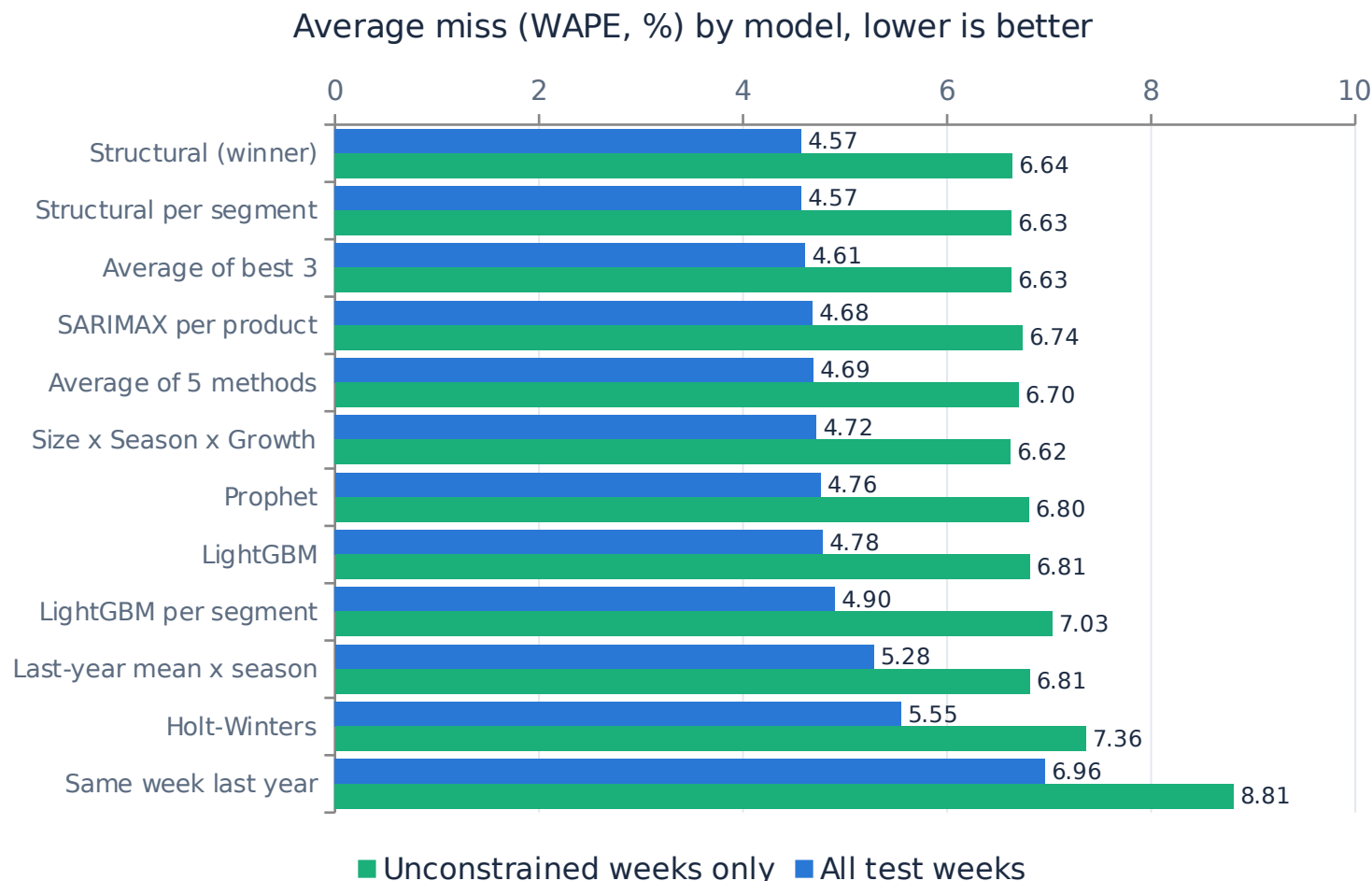
Worked example: Barley, week 235 (first test week)

2,579	× 1.057	× 0.983	× 1.002	= 2,687	2,505
level	trend	season	small fix	forecast	actual



Barley, 26 test weeks: average miss 4.8% (all products 4.6%). Free parameters (seasonal detail K = 3, one shared trend) were picked on validation weeks inside the training data only; every competing model was tuned the same way.

13 models compared: the simple structure wins

**4.57%**

winner, all test weeks

6.64%

winner, unconstrained weeks

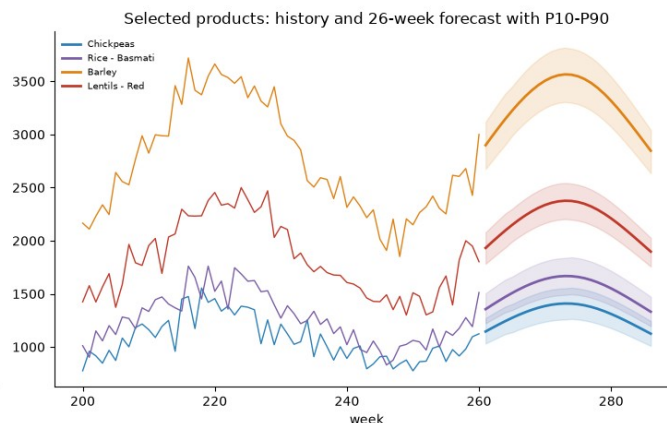
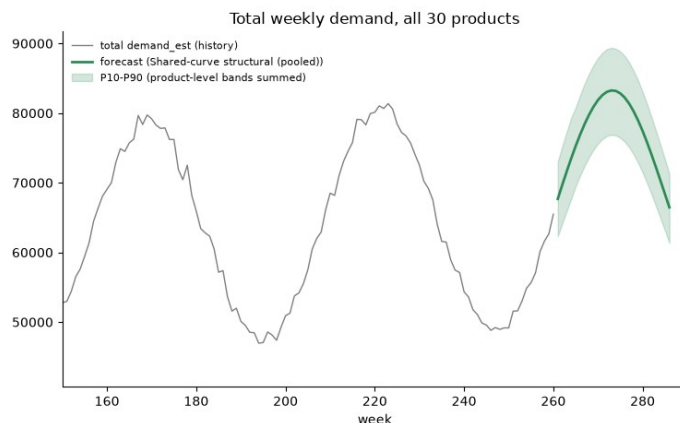
Honest figure: 6.6% where demand was observed directly.

Top models tie there: 6.6 to 6.8%. The structure matters, not complexity.

ML does not win: LightGBM 4.78%, Prophet 4.76%.

13% better than the best simple rule, 34% than last year's week.

Next 26 weeks: \$64.6M potential, \$28.7M unserved



2.0M
units
forecast

+2.8%
vs last year

80%
inside P10-
P90

Range P10-P90 from backtest errors: 1.84M to 2.15M units

1,998k units

forecast demand (P50), summed over 780 product-weeks

× last-13-week price

each product's own average price

= \$64.6M

revenue potential if every unit could be sold

stock simulation

sold = min(demand, stock drawn from the last 104 weeks): 1,093k units

= \$35.8M

expected sales at historical stock levels

\$64.6M - \$35.8M = \$28.7M

unserved demand (44%)

Example, Barley week 1: 2,899 units × \$7.447 = \$21,589 potential; history lets 1,397 units sell (\$10,406); 1,494 units (\$11,182) unserved. Assumptions: stock follows each product's history; prices stay at the last 13-week average.

KPI dashboard: financial, operational, credit

Commercial performance dashboard

Armani Middle East Trading | 30 food commodities | what happened, what to expect, what to do | dates are a synthetic calendar

Overview Financial Stock & service Customer credit (scenario) Forecast Pricing Alerts Notes

Bottom line: margins are stable (21.5% gross), but **39.8% of customer demand went unserved** because products ran out of stock (about \$43.2M of sales), while 5 products hold far more stock than they sell. The opportunity is availability and stock planning, not price cuts.

Revenue ⓘ \$65.4M ↑ +3.3% vs last year	Gross profit margin ⓘ 21.5% ↓ -0.1 pp	Operating profit ⓘ \$7.6M ↑ +2.8%	Demand served (service level) ⓘ 60.0% ↑ +1.1 pp
Stock turnover (times per year) ⓘ 3.9 ↓ -14.8%	Days of stock (DSI) ⓘ 92 days ↑ +17.4%	DSO - scenario ⓘ 37 days	Default risk rating - scenario ⓘ D ↑ worst: Beverages & Cocoa

What the numbers say

- **Service level 60.0%:** we served this share of what customers wanted in last 52 weeks.
- **5 products hold more than 26 weeks of stock**, tying up roughly \$10.9M at cost, while the lean products run out.
- **Next 26 weeks:** demand is forecast at 1,997,953 units (+2.8% vs the same weeks last year); at historical stock levels about **44.4%** of it would again be unserved.

Overview tab (live demo)

\$65.4M

revenue, +3.3% vs last year

21.5%

gross margin, flat

60%

of demand served; \$43.2M lost

16 high

priority alerts, each with an action

Financial

gross margin · revenue growth · operating profit

Operational

inventory turnover · days of stock (DSI) · service level · lost revenue

Credit

days sales outstanding (DSO) · default-risk rating A-E (scenario)

Forward view and actions

26-week forecast with P10-P90 range · price plan · six rule-based alerts

Credit-risk rating A to E: what, how, and why

Risk score (0-100) = weighted sum of four components

40%

Default rate

annual default of credit sales; 3% = score 100

20%

Lateness

average delay vs terms; delay = half the terms = 100

25%

Loss vs margin

expected credit loss as share of gross profit; 15% = 100

15%

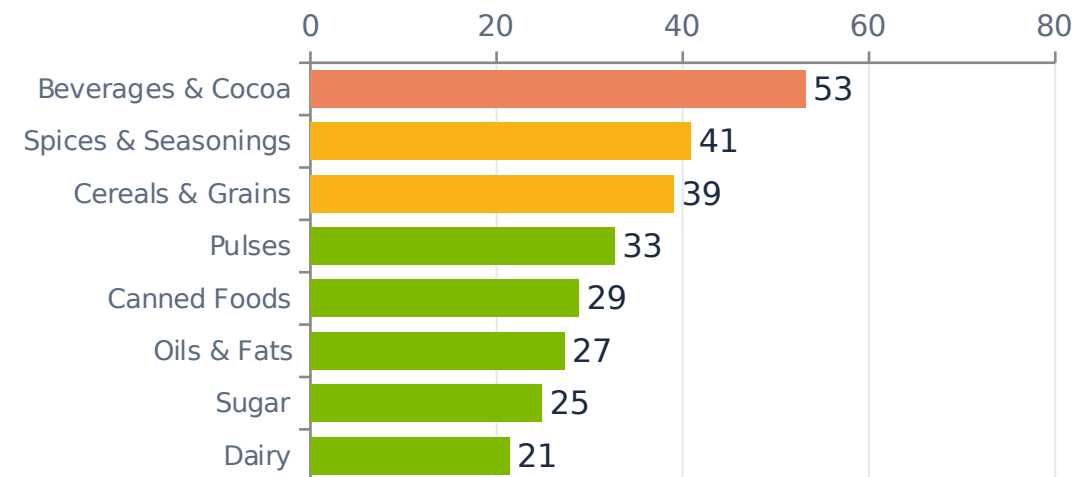
Concentration

category share of revenue; 25% = 100

Why it matters

- Raises credit alerts in the dashboard.
- Sets pricing rules: C markdown $\leq 5\%$; D/E no markdown, risk premium, 15 days shorter terms.

Risk score by category (scenario inputs)



A

< 20

B

< 35

C

< 50

D

< 65

E

≥ 65

Beverages & Cocoa: $0.40 \times 60 + 0.20 \times 40 + 0.25 \times 42 + 0.15 \times 71 = 53 \rightarrow D$

Inputs are scenario assumptions (no customer data): terms, share on credit, delay, default rate per category.

Four data insights point to one pricing idea

INSIGHT 1

78%

of forecast product-weeks are scarce: stock cannot cover demand

INSIGHT 2

±25%

yearly demand swing: peak week 13, quiet week 38

INSIGHT 3

\$10.8M

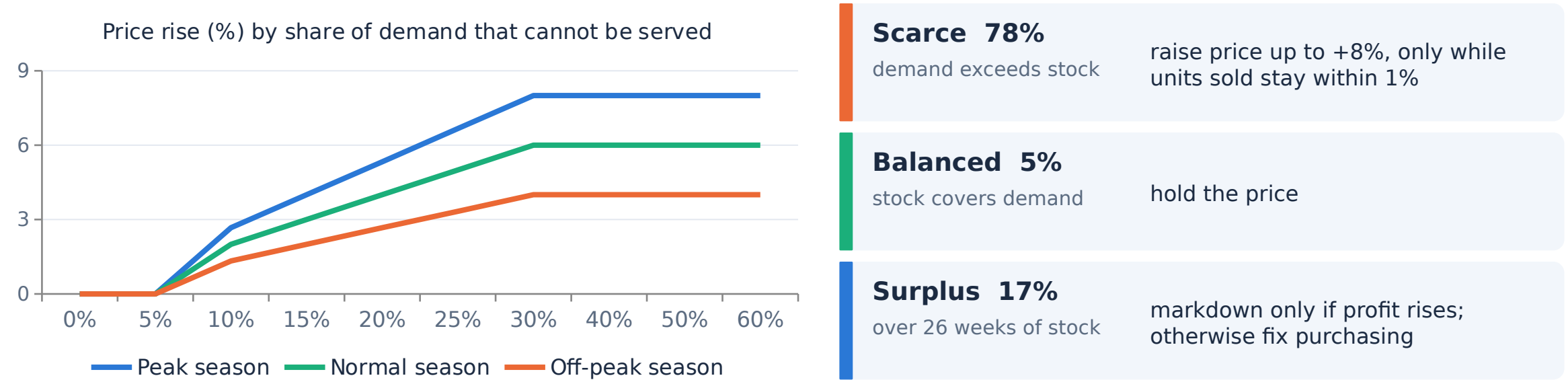
tied up in 5 over-stocked products (64 to 370 weeks of stock)

INSIGHT 4

≈ 0

measurable price effect: -0.017 (range -0.056 to +0.021)

So the idea: raise price where stock sells out anyway · ignore unknown price sensitivity · discount only if it pays



From forecast to price in five steps

INPUTS



OUTPUT recommended price for every product-week, with the reason

Example, Barley week 1: \$7.447 → \$7.67 (+3.0%) · scarce, only 48% of demand can be served · limited by the weekly step

Price follows the forecast peaks and lulls



Rise = 8% × min(1, g / 30%) × season factor

g = share of expected demand that cannot be served

Unserved demand g	Peak ×1.0	Normal ×0.75	Off-peak ×0.5
10%	+2.7%	+2.0%	+1.3%
20%	+5.3%	+4.0%	+2.7%
30% or more	+8.0%	+6.0%	+4.0%
5% or less	0%	0%	0%

How to read the Barley paths

- Peak = shared seasonal curve at or above 1.10; off-peak at or below 0.90.
- Weeks 1-3: normal season, the 3% weekly step limit slows the first rise.
- Weeks 5-22: peak, price sits at the +8% cap.
- Weeks 23-26: season turns, the premium eases toward +4%.

Pricing rules: lean vs high-stock, by season

	Peak season	Normal season	Off-peak season
Lean products stock below demand · 78% of product-weeks	up to +8% full scarcity premium	up to +6% 75% of the premium	up to +4% 50% of the premium
High-stock products over 26 weeks of cover · 17% · 5 products	hold the price no markdown in a peak	markdown ≤ 5% only if profit rises	markdown ≤ 10% only if profit rises

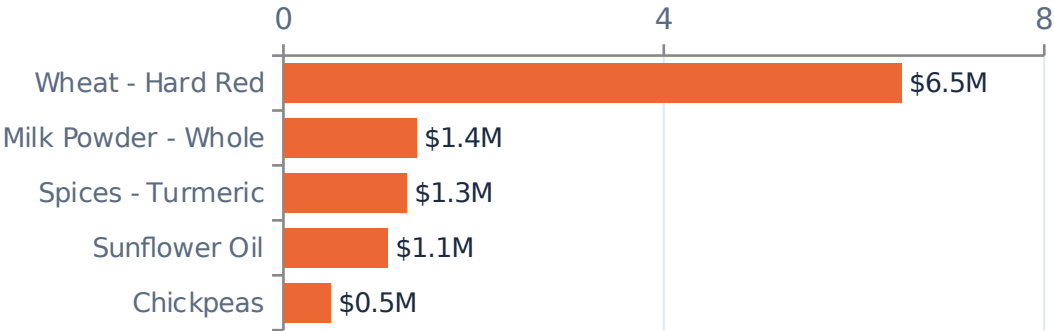
Lean products

Stock, not price, limits sales, so the premium is free money. It is largest in a peak and smallest in a lull.

High-stock products

Price cannot fix 64-370 weeks of stock: clearing it in a year needs cuts of 17-49%. Stop buying; offer bulk or export deals. Carrying cost \$1.3M a year exceeds the whole pricing gain.

Capital tied up above 26 weeks of stock (at cost)

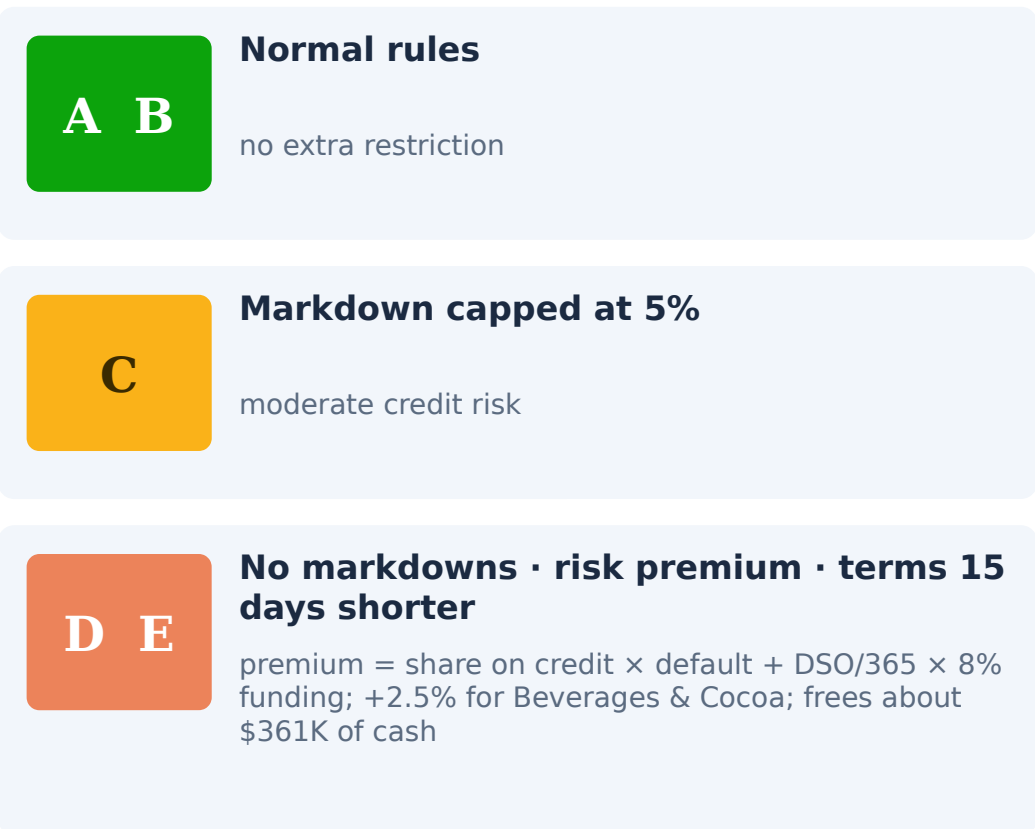


Guardrails: competitive, safe and credit-aware

Competitive and cost corridor (Barley, \$ per unit)

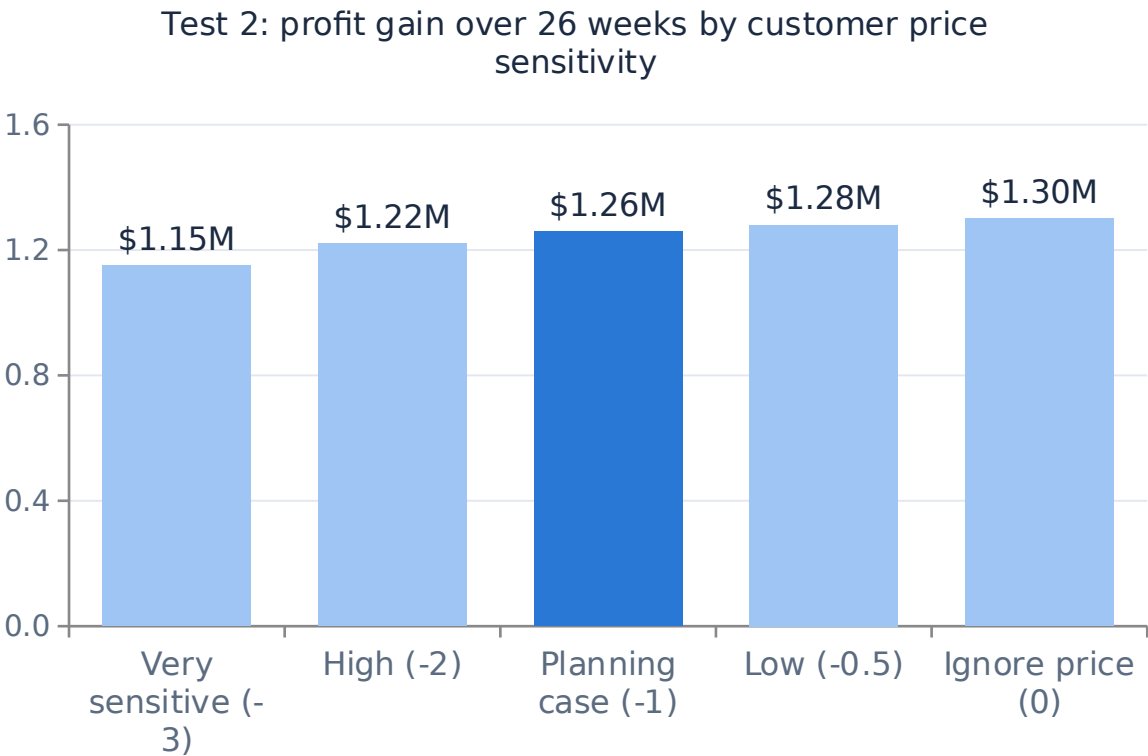


Credit-risk guardrails by rating



All 19 automatic checks pass: band, floor, weekly step, credit rules and hand-calculated arithmetic.

Tested four ways: gain stays near +\$1.3M



1 Simulate

1,000 demand and stock draws per product-week; same draws with and without the engine

2 Stress sensitivity

price sensitivity 0 to -3: result is a range

3 Replay the past

last 52 weeks, only past information: +\$1.43M

4 Stress the inputs

16 settings all positive; demand at 60% of estimate: still +\$0.94M

19 / 19

automatic checks pass

-0.017

price effect: not measurable, hence a range

Next: two 8-week pilots. A: 6 scarce products at +3% / +5% vs matched controls (do they still sell out?). B: high-stock products at $\pm 3\%$ in random weeks (real price sensitivity).

Thanks for your attention

Questions are welcome

Hamidreza Nademi | Technical Assessment for Data Scientist Position